

Lesson 5 Lesson-End Project

Designing Solution for Logging and Monitoring

Project agenda: To implement Azure Monitor alerts for the given scenario

Description: You have been given a project to design monitoring for a web application. You should be alerted whenever the number of HTTP 404 errors for the web app crosses a threshold of 5. You also need to ensure that appropriate stakeholders are notified about it.

Tools required: Azure account with administrator access

Prerequisites: None

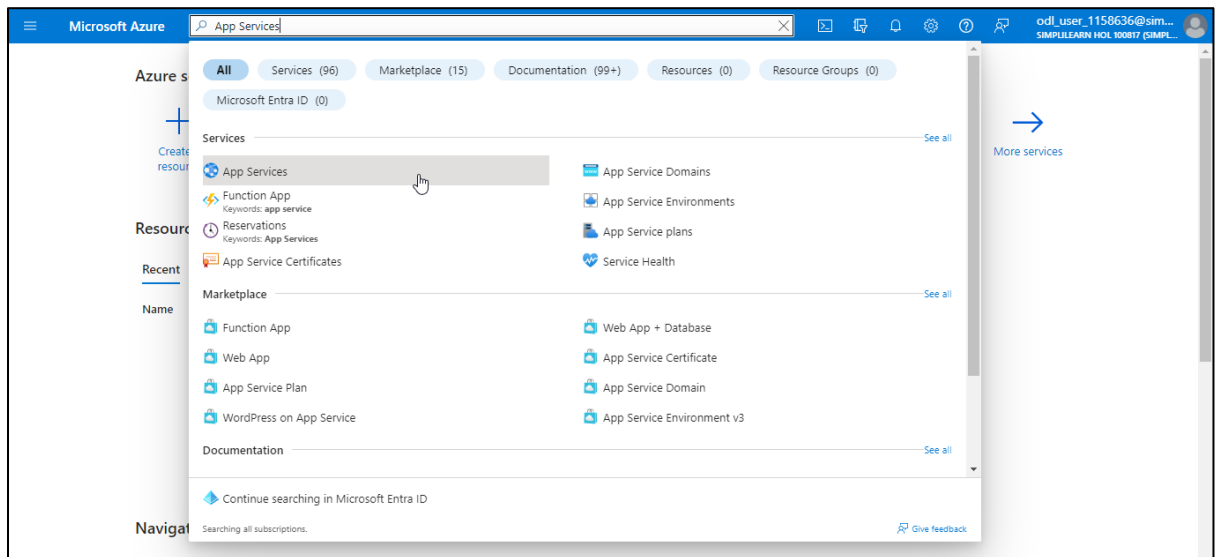
Expected deliverables: Illustrate the solution to the given scenario, create alerts as mentioned, and add screenshots containing the respective outputs

Steps to be followed:

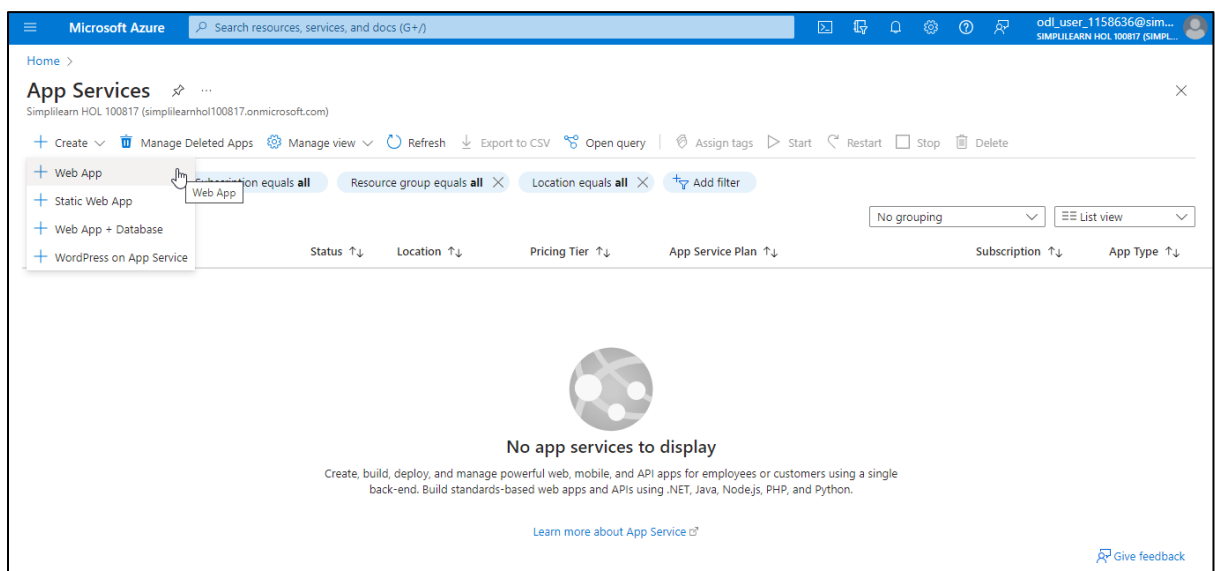
1. Create a web app
2. Create an Azure alert

Step 1: Create a web app

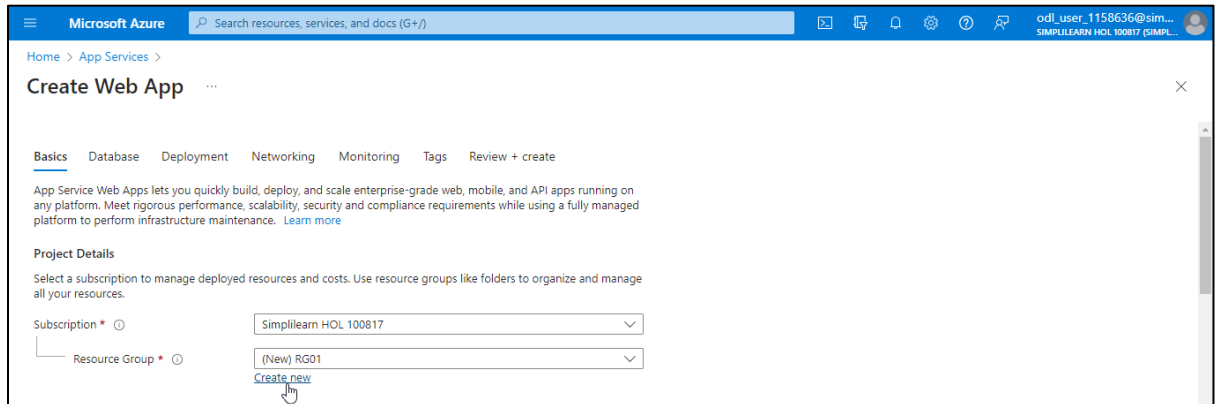
1.1 Log in to the Azure portal, search for and select **App Services**



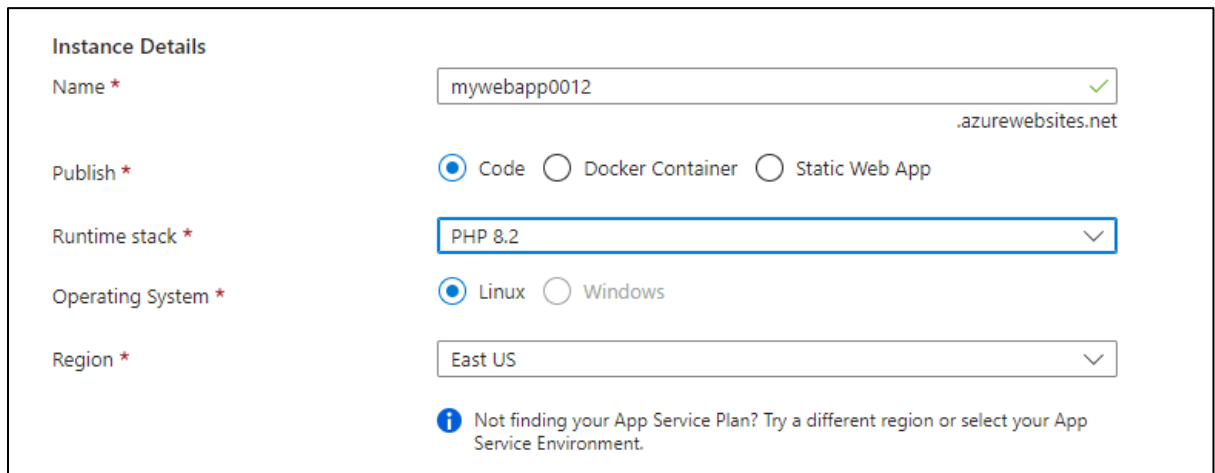
1.2 On the **App Services** page, click + **Create** and select + **Web App**



1.3 On the **Create Web App** page, navigate to the **Basics** tab. Now, under **Project Details**, click on **Create new** to create a new **Resource Group (RG01)**



1.4 Under **Instance Details**, provide the information as shown in the screenshot below:



The screenshot shows the 'Instance Details' form for creating a new App Service Web App. The fields are as follows:

- Name ***: mywebapp0012 (with a green checkmark and the domain .azurewebsites.net)
- Publish ***: Code (selected), Docker Container, Static Web App
- Runtime stack ***: PHP 8.2
- Operating System ***: Linux (selected), Windows
- Region ***: East US

Below the form, there is an information icon and the text: "Not finding your App Service Plan? Try a different region or select your App Service Environment."

1.5 Under **Pricing plans**, select **Free F1 (Shared infrastructure)**

Pricing plans

App Service plan pricing tier determines the location, features, cost and compute resources associated with your app.
[Learn more](#)

Linux Plan (East US) * ⓘ (New) ASP-RG01-99ae
[Create new](#)

Pricing plan
 Free F1 (Shared infrastructure)

Zone redundancy

An App Service plan can be deployed as a zone-redundant or a single-zone plan. You can't make an App Service plan zone-redundant after it's been created.

Zone redundancy

Popular plans

- Free F1
60 CPU Minutes/day included
- Basic B1
ACU: 100, Memory: 1.75 GB, vCPU: 1
- Premium V3 P1V3
ACU: 195, Memory: 8 GB, vCPU: 2
- Premium V3 P2V3
ACU: 195, Memory: 16 GB, vCPU: 4
- Premium V3 P3V3
ACU: 195, Memory: 32 GB, vCPU: 8

[Review + create](#) [< Previous](#) [Next : Database >](#)

1.6 Click **Review + create**

Microsoft Azure Search resources, services, and docs (G+)

Home > App Services >

Create Web App

Basics Database Deployment Networking Monitoring Tags Review + create

App Service Web Apps lets you quickly build, deploy, and scale enterprise-grade web, mobile, and API apps running on any platform. Meet rigorous performance, scalability, security and compliance requirements while using a fully managed platform to perform infrastructure maintenance. [Learn more](#)

Project Details

Select a subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Simplilearn HOL 100817

Resource Group * ⓘ (New) RG01
[Create new](#)

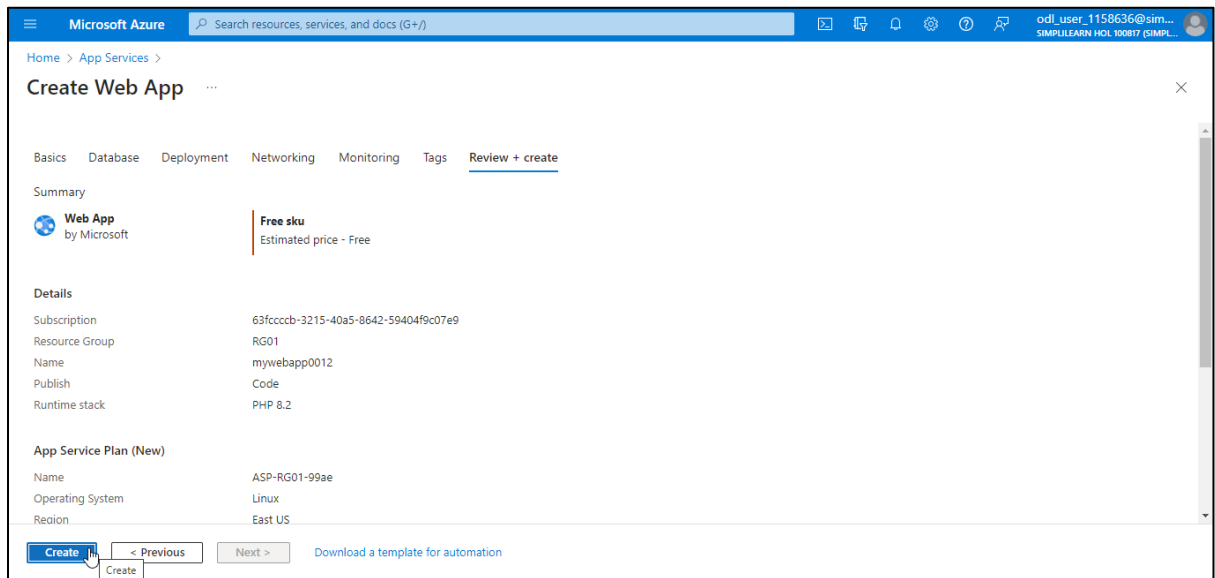
Instance Details

Name * mywebapp0012 ✓
 .azurewebsites.net

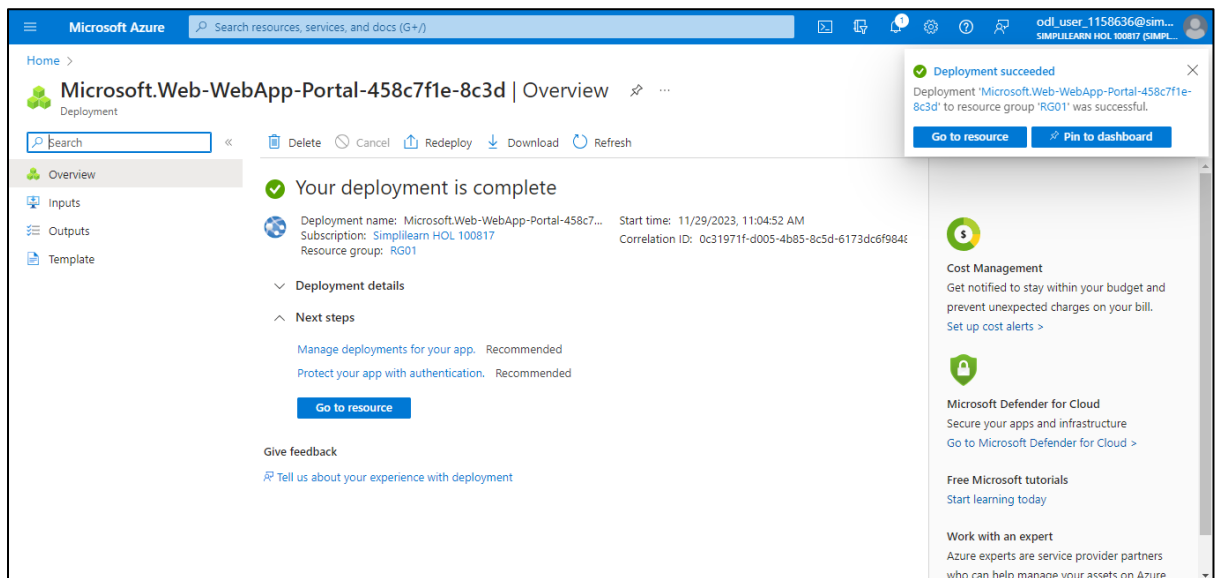
Publish * ☒ Code ☐ Docker Container ☐ Static Web App

[Review + create](#) [< Previous](#) [Next : Database >](#)

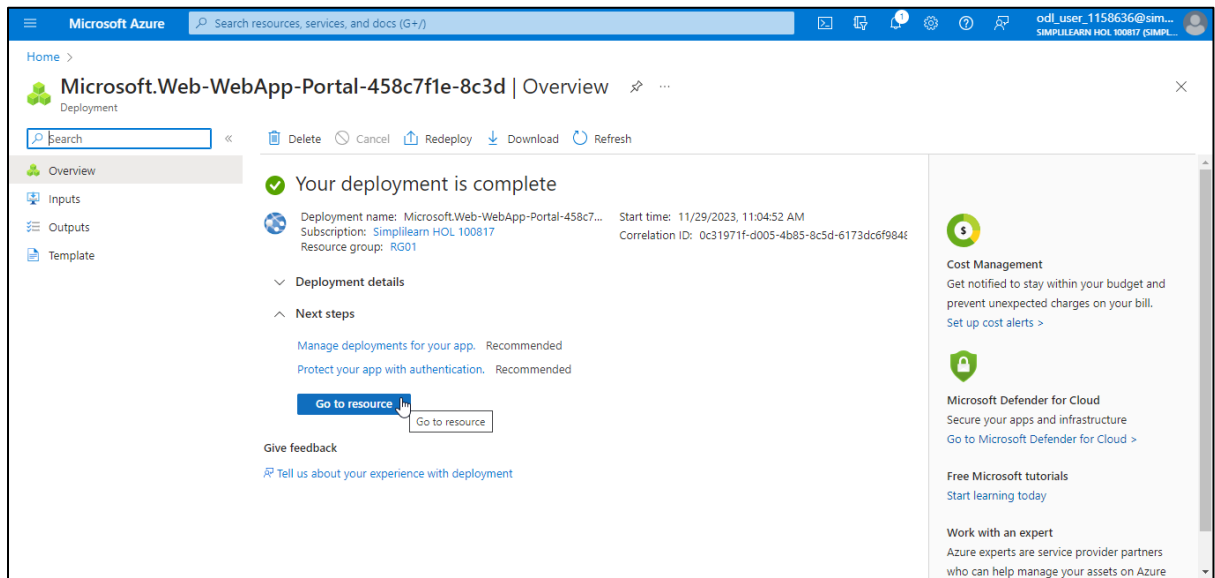
1.7 Click Create



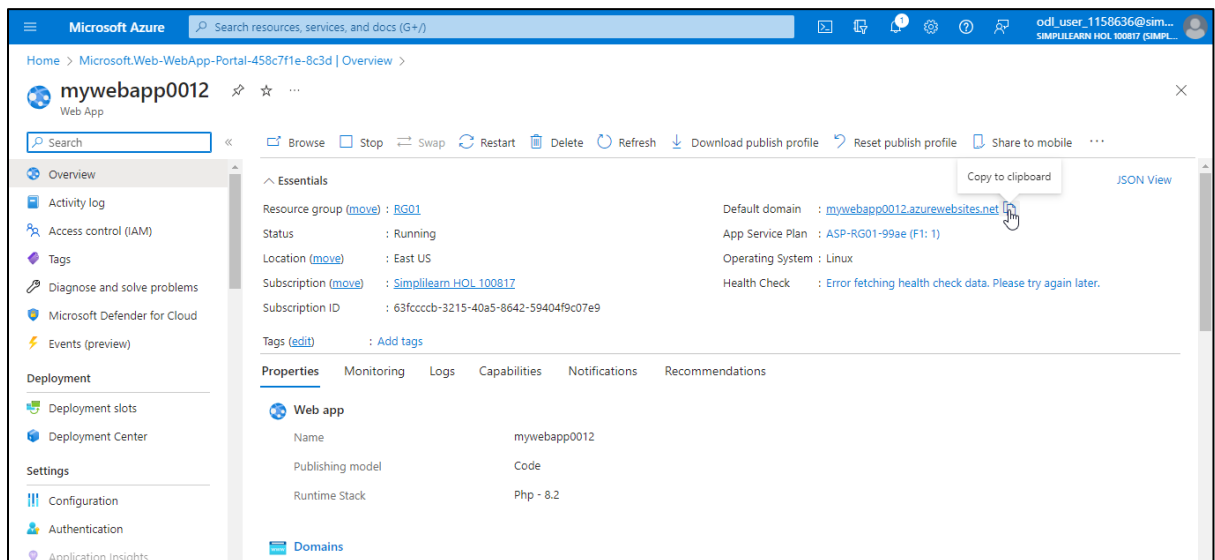
The web app has been created successfully as shown in the screenshot below:



1.8 Click **Go to resource**

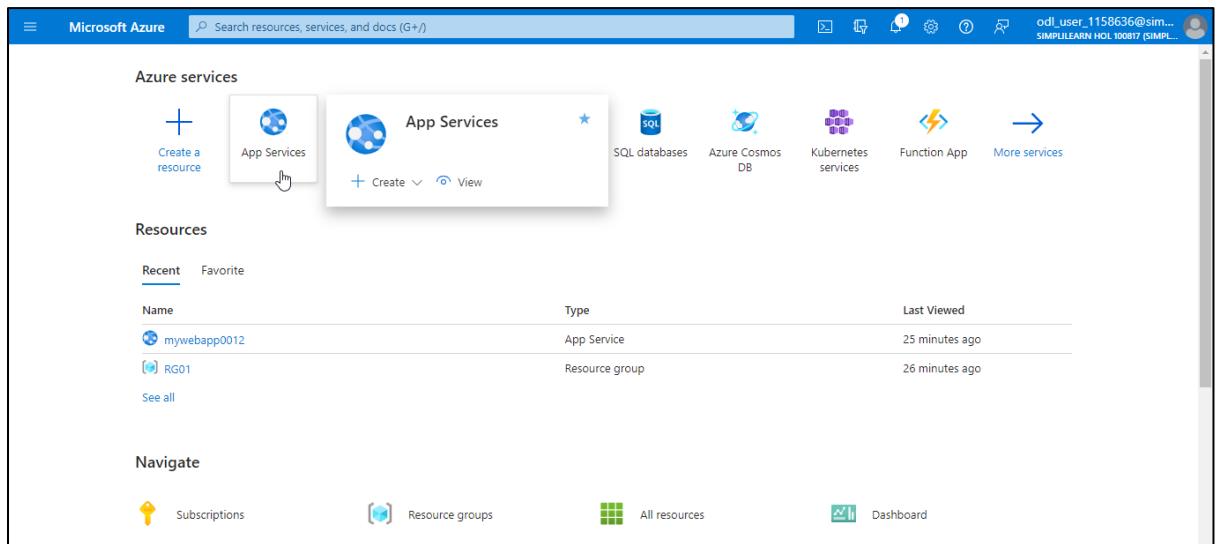


On the **mywebapp0012** page, you can view the URL **mywebapp0012.azurewebsites.net** for the web app created under the **Default domain**:

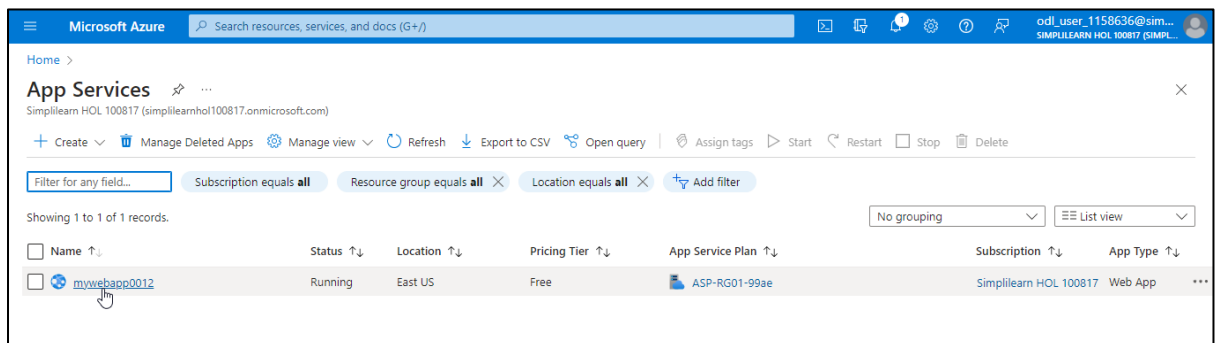


Step 2: Create an Azure alert

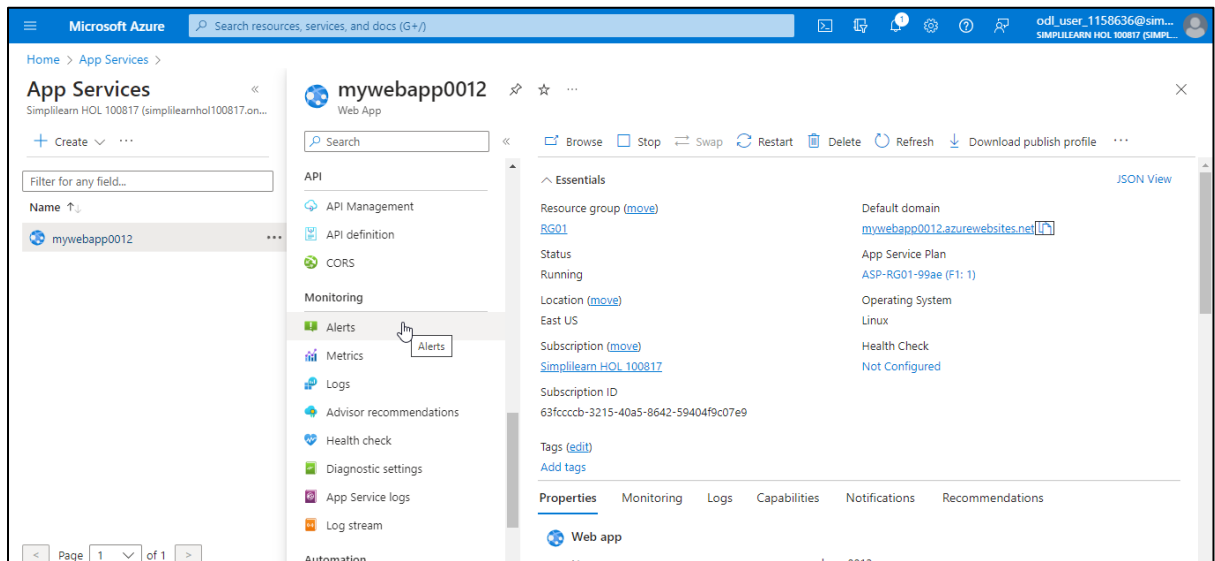
2.1 Navigate back to the Microsoft Azure home page and select App Services



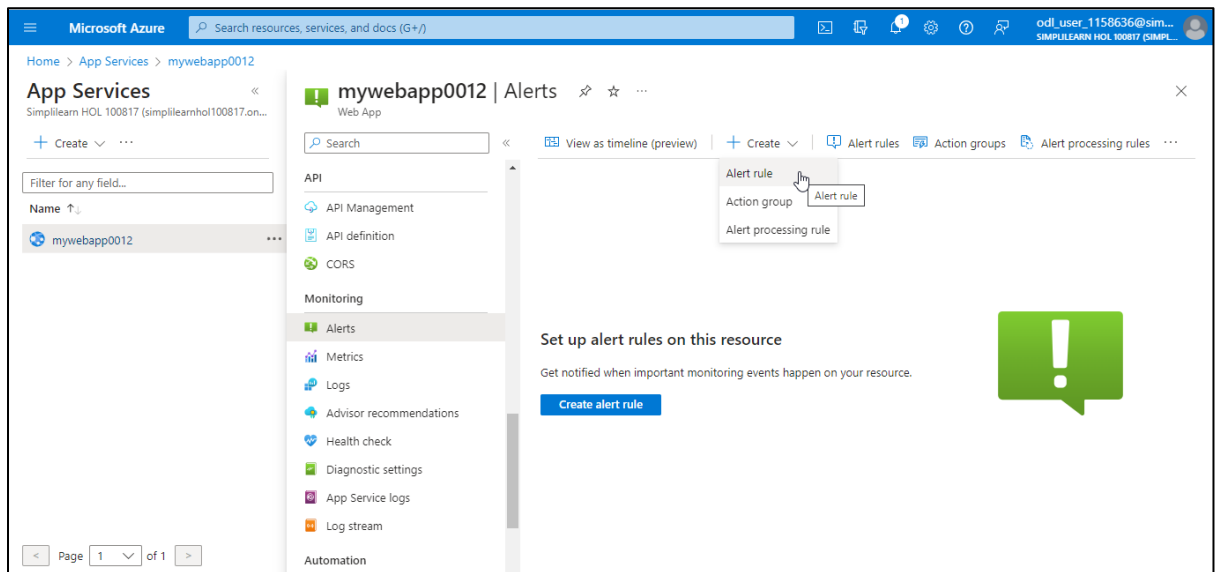
2.2 In the App Services page, click on the web app (mywebapp0012) that was created earlier



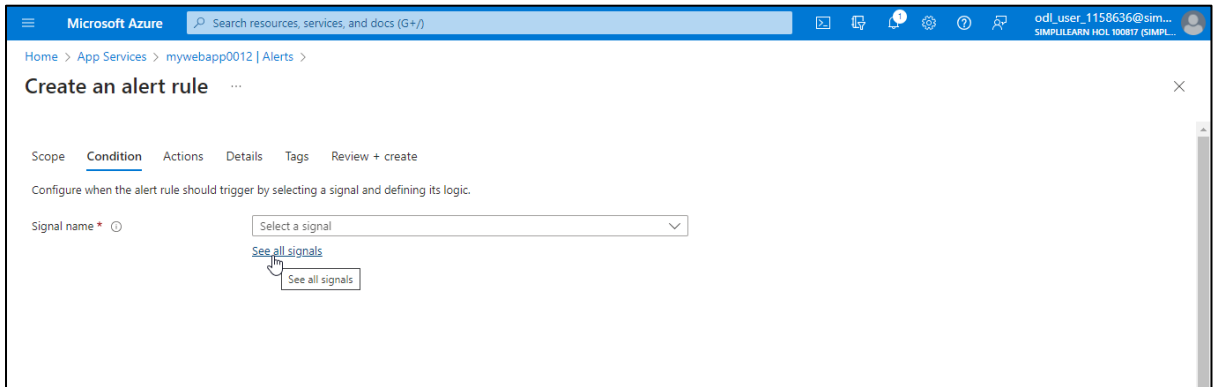
2.3 On the **mywebapp0012** page, click on **Alerts** under the **Monitoring** section



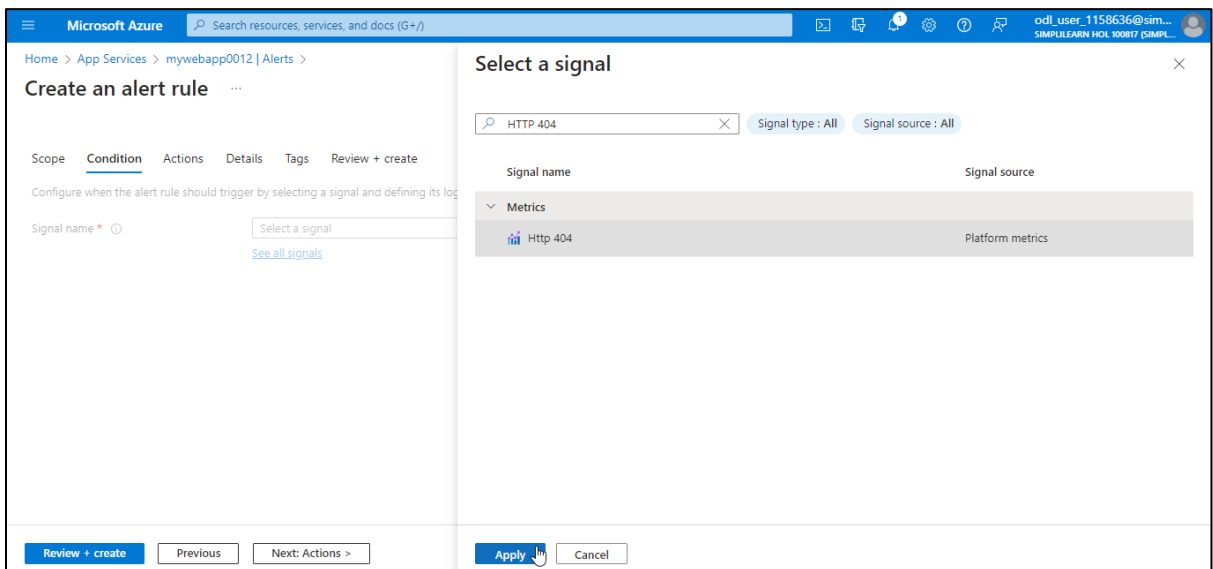
2.4 On the **Alerts** page, click on **+ Create** and select **Alert rule**



2.5 On the **Create an alert rule** page, navigate to the **Condition** tab and click on **See all signals** under the **Signal name**



2.6 On the **Select a signal** page, search for and select **HTTP 404** and click **Apply**



2.7 Provide the other details as shown in the screenshot below and click **Next: Actions >**

The screenshot shows the 'Create an alert rule' page in the Microsoft Azure portal. The breadcrumb navigation is 'Home > App Services > mywebapp0012 | Alerts >'. The page title is 'Create an alert rule'. The tabs are 'Scope', 'Condition', 'Actions', 'Details', 'Tags', and 'Review + create'. The 'Condition' tab is active. The instructions say 'Configure when the alert rule should trigger by selecting a signal and defining its logic.' The 'Signal name' is 'Http 404'. The 'Alert logic' section has 'Static' selected for 'Alert logic', 'Total' for 'Aggregation type', 'Greater than' for 'Operator', 'Count' for 'Unit', and '5' for 'Threshold value'. A 'Preview' section on the right shows a graph with a peak at 5 and a cost of '\$0.10 USD/month'. The bottom navigation bar has 'Review + create', 'Previous', 'Next: Actions >', and 'Next: Actions' buttons.

2.8 In the **Actions** tab, click on **+ Create action group**

The screenshot shows the 'Create an alert rule' page in the Microsoft Azure portal, with the 'Actions' tab active. The breadcrumb navigation is 'Home > App Services > mywebapp0012 | Alerts >'. The page title is 'Create an alert rule'. The tabs are 'Scope', 'Condition', 'Actions', 'Details', 'Tags', and 'Review + create'. The 'Actions' tab is active. The instructions say 'An action group is a set of actions that can be applied to an alert rule. [Learn more](#)'. There are two buttons: '+ Select action groups' and '+ Create action group'. Below them is a table with columns 'Action group name' and 'Contains actions'. The table is empty, with the text 'No action group selected yet' below it. The bottom navigation bar has 'Review + create', 'Previous', and 'Next: Details >' buttons.

2.9 In the **Create action group** page, navigate to the **Basics** tab and provide the information under **Project details** as shown below:

The screenshot shows the 'Create action group' page in the Microsoft Azure portal. The 'Basics' tab is selected. Under the 'Project details' section, the following information is entered:

- Subscription: Simplilearn HOL 100817
- Resource group: RG01
- Region: Global

There is a 'Create new' link next to the Resource group dropdown.

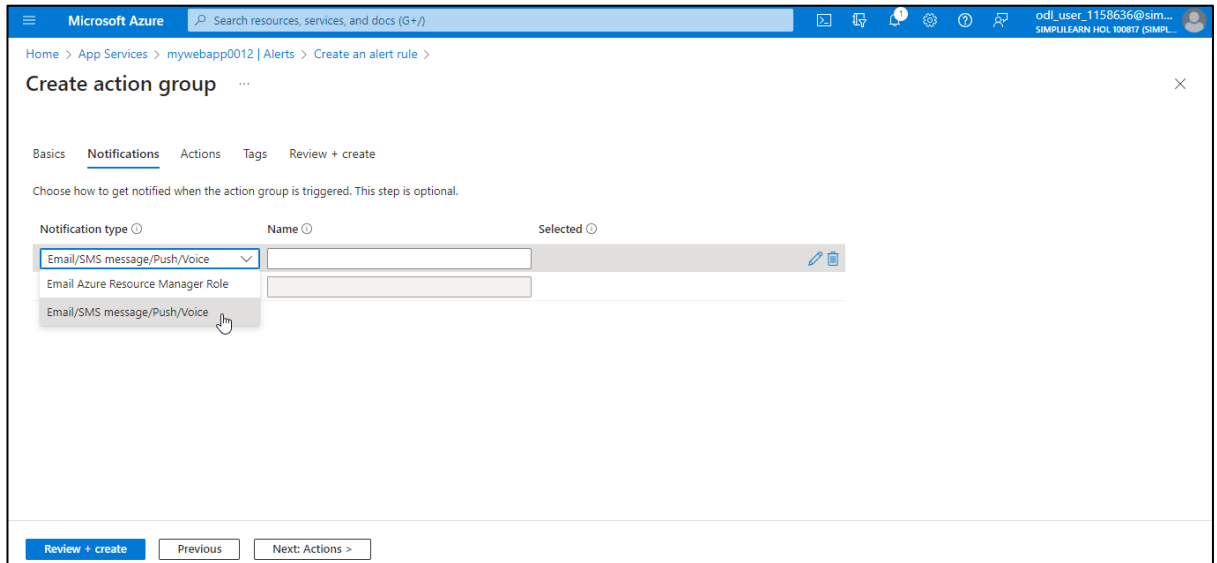
2.10 Under **Instance details**, provide the information as shown in the screenshot below and click **Next: Notifications >**

The screenshot shows the 'Create action group' page in the Microsoft Azure portal, with the 'Instance details' section visible. The following information is entered:

- Action group name: actiongroup1
- Display name: actiongroup1

At the bottom of the page, there are three buttons: 'Review + create', 'Previous', and 'Next: Notifications >'. A tooltip is visible over the 'Next: Notifications >' button, showing the text 'Next:Notifications'.

2.11 In the **Notifications** tab, choose **Email/SMS message/Push/Voice** from the drop-down under **Notification type**



Microsoft Azure Search resources, services, and docs (G+/)

Home > App Services > mywebapp0012 | Alerts > Create an alert rule >

Create action group

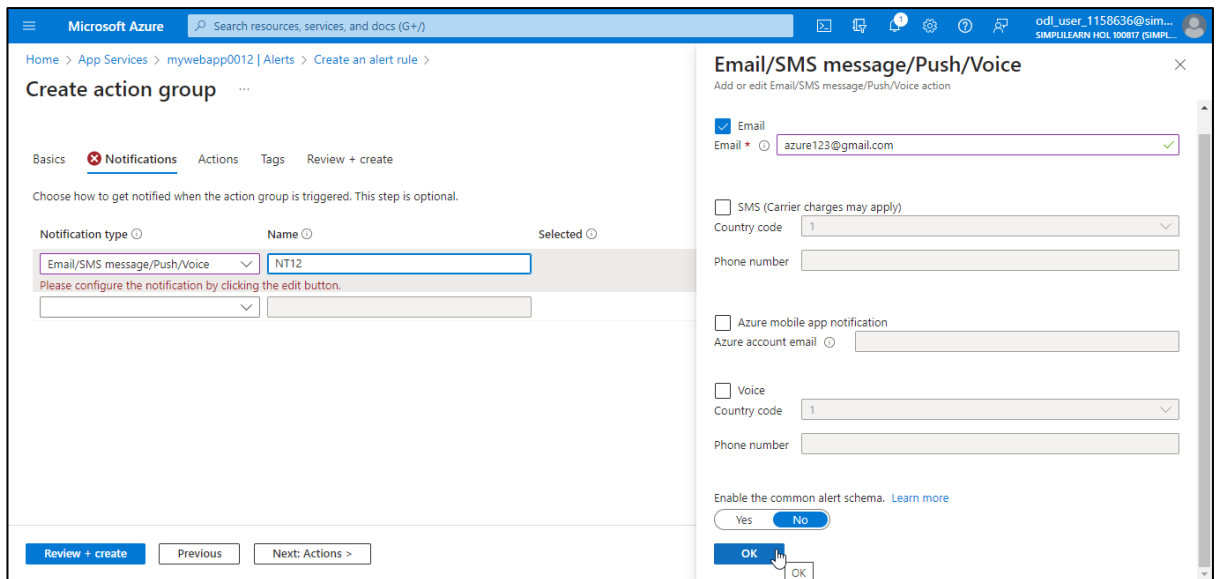
Basics **Notifications** Actions Tags Review + create

Choose how to get notified when the action group is triggered. This step is optional.

Notification type	Name	Selected
Email/SMS message/Push/Voice		
Email Azure Resource Manager Role		
Email/SMS message/Push/Voice		

Review + create Previous Next: Actions >

2.12 In the **Email/SMS message/Push/Voice** page, provide the details as shown below and click **OK**



Microsoft Azure Search resources, services, and docs (G+/)

Home > App Services > mywebapp0012 | Alerts > Create an alert rule >

Create action group

Basics **Notifications** Actions Tags Review + create

Choose how to get notified when the action group is triggered. This step is optional.

Notification type	Name	Selected
Email/SMS message/Push/Voice	NT12	

Please configure the notification by clicking the edit button.

Email/SMS message/Push/Voice

Add or edit Email/SMS message/Push/Voice action

☒ Email

Email * azure123@gmail.com ✓

☐ SMS (Carrier charges may apply)

Country code 1

Phone number

☐ Azure mobile app notification

Azure account email

☐ Voice

Country code 1

Phone number

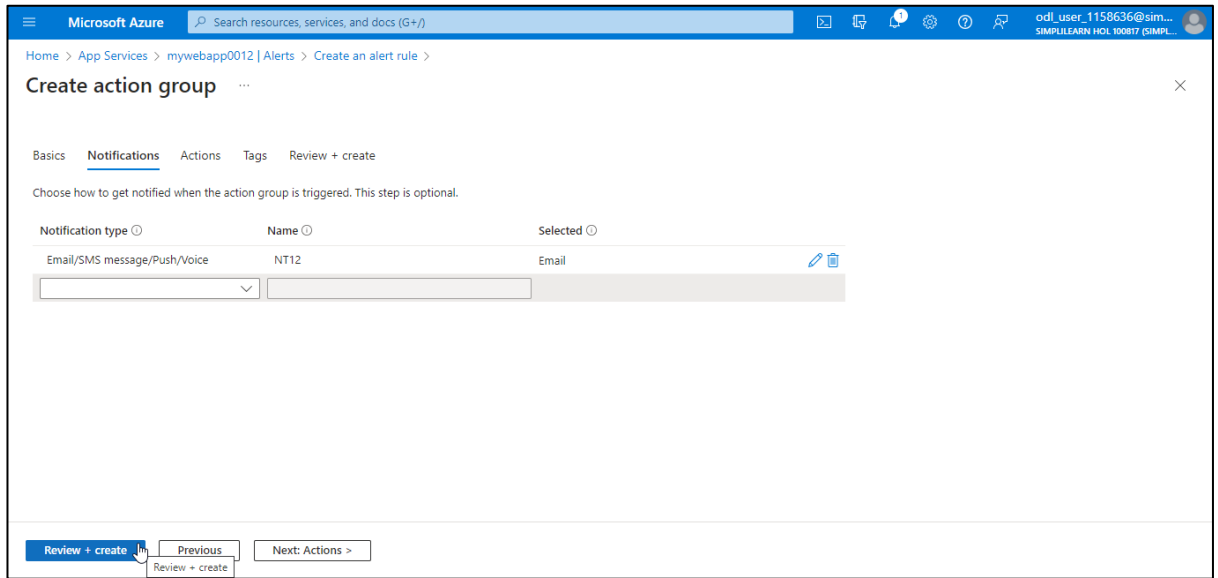
Enable the common alert schema. [Learn more](#)

Yes No

OK OK

Review + create Previous Next: Actions >

2.13 Provide a name under the **Name** section and click **Review + create**



Microsoft Azure Search resources, services, and docs (G+)

Home > App Services > mywebapp0012 | Alerts > Create an alert rule >

Create action group

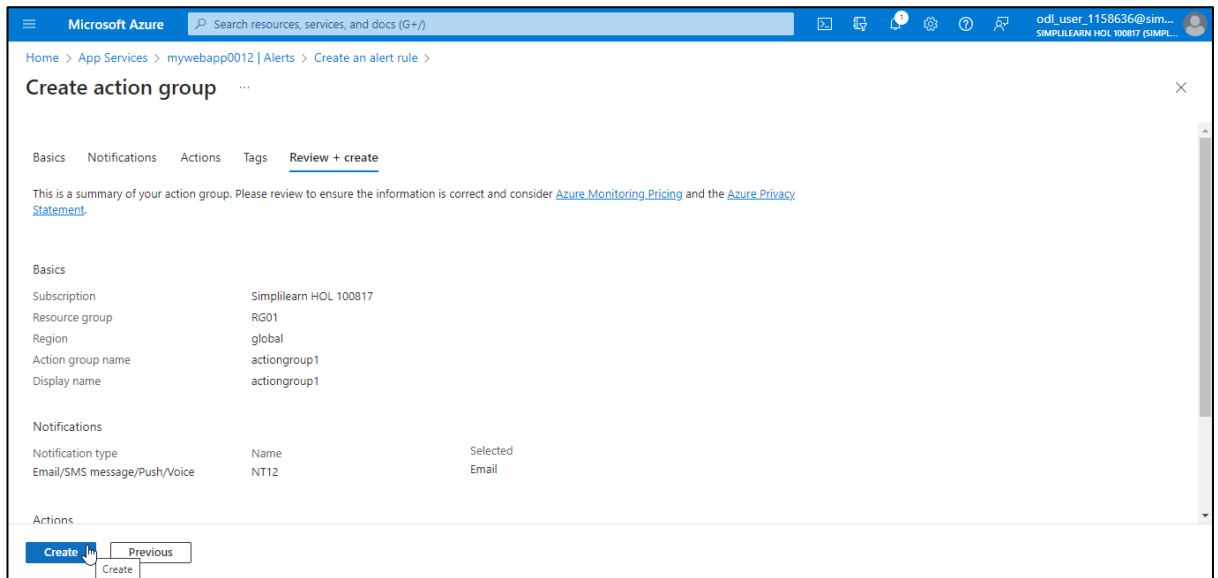
Basics **Notifications** Actions Tags Review + create

Choose how to get notified when the action group is triggered. This step is optional.

Notification type	Name	Selected
Email/SMS message/Push/Voice	NT12	Email

Review + create Previous Next: Actions >

2.14 Click **Create**



Microsoft Azure Search resources, services, and docs (G+)

Home > App Services > mywebapp0012 | Alerts > Create an alert rule >

Create action group

Basics Notifications Actions Tags **Review + create**

This is a summary of your action group. Please review to ensure the information is correct and consider [Azure Monitoring Pricing](#) and the [Azure Privacy Statement](#).

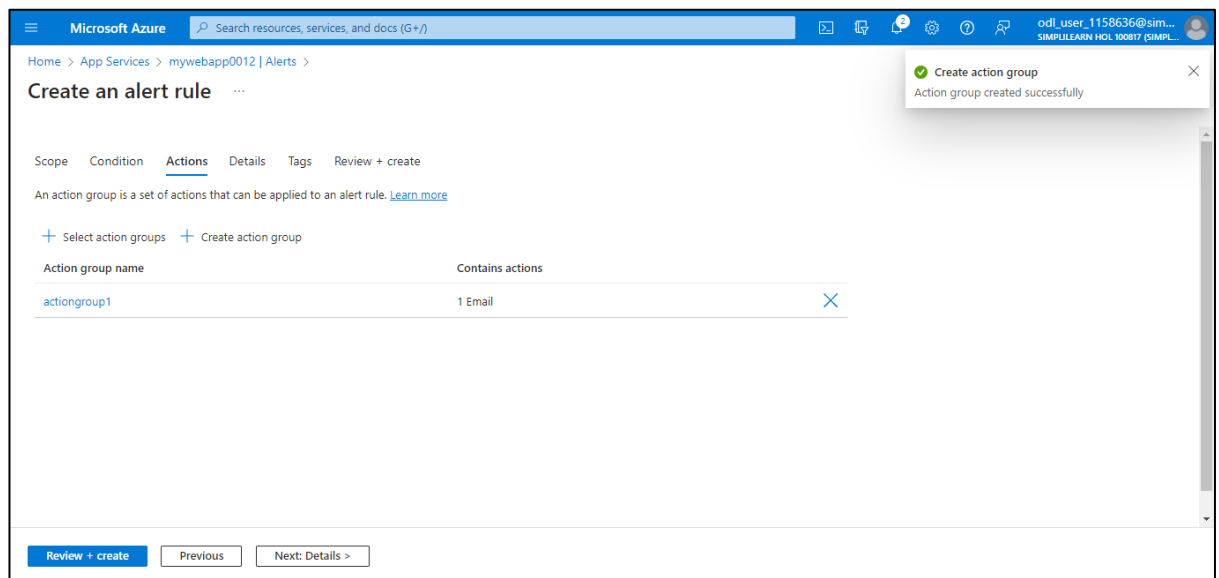
Basics	
Subscription	Simplilearn HOL 100817
Resource group	RG01
Region	global
Action group name	actiongroup1
Display name	actiongroup1

Notifications		
Notification type	Name	Selected
Email/SMS message/Push/Voice	NT12	Email

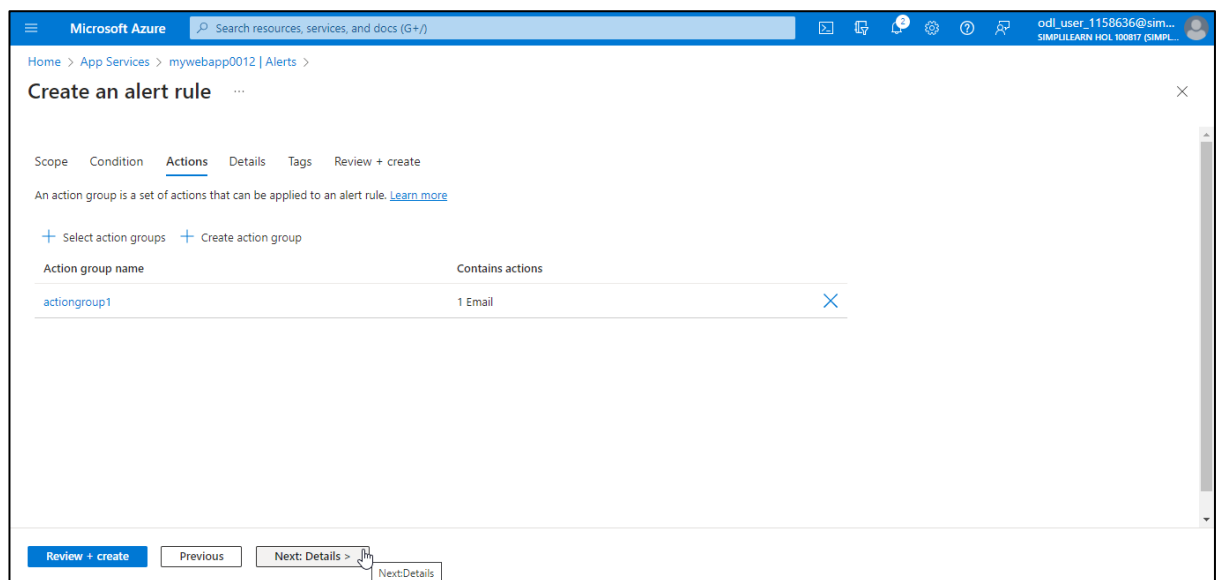
Actions

Create Previous

The action group (**actiongroup1**) is created as shown below:



2.15 Click **Next: Details >**



2.16 In the **Details** tab, provide the information as shown below and click **Review + create**

The screenshot shows the 'Create an alert rule' form in the 'Details' tab. The form is titled 'Create an alert rule' and has tabs for 'Scope', 'Condition', 'Actions', 'Details', 'Tags', and 'Review + create'. The 'Details' tab is active.

Project details

Select the subscription and resource group in which to save the alert rule.

Subscription *

Resource group * [Create new](#)

Alert rule details

Severity *

Alert rule name * ✓

Alert rule description

At the bottom, there are three buttons: 'Review + create' (highlighted with a mouse cursor), 'Previous', and 'Next: Tags >'. A tooltip for the 'Review + create' button shows the text 'Review + create'.

2.17 Click **Create**

The screenshot shows the 'Create an alert rule' form in the 'Review + create' tab. The form is titled 'Create an alert rule' and has tabs for 'Scope', 'Condition', 'Actions', 'Details', 'Tags', and 'Review + create'. The 'Review + create' tab is active.

Metric alert rule

1 Condition [Terms of use](#) | [Privacy statement](#)

Total pricing
0.10 USD/month
[Pricing](#)

Scope

Resource [Simplilearn HOL 100817](#) > [RG01](#) > [mywebapp0012](#)

Condition

Signal name [Http404](#)

Operator [Greater than](#)

Aggregation type [Total](#)

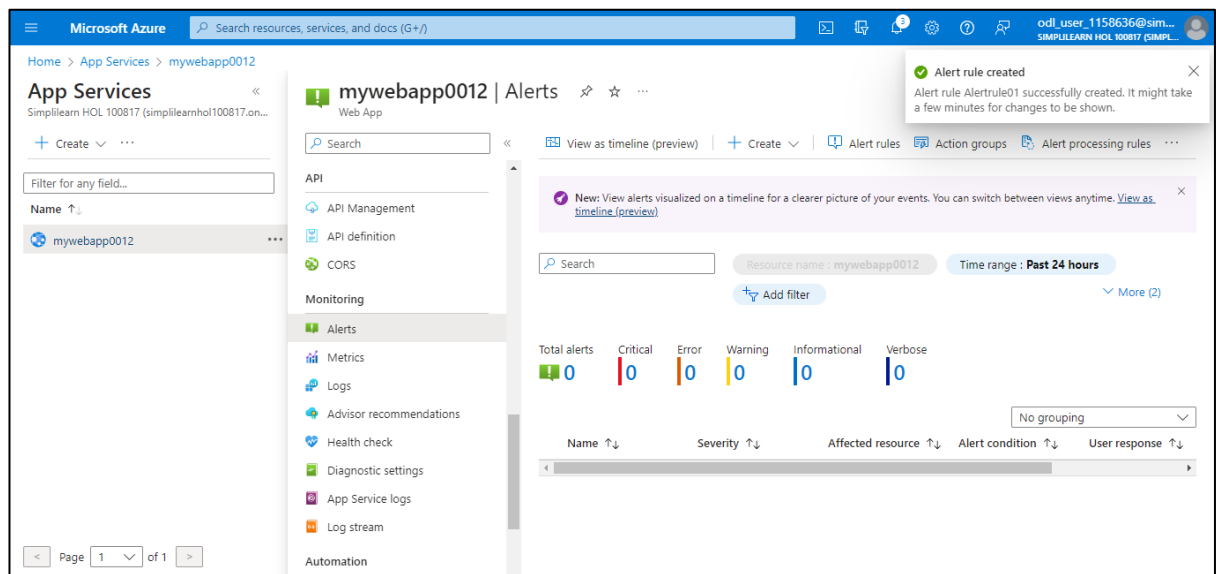
Threshold value [5](#)

Lookback period [5 minutes](#)

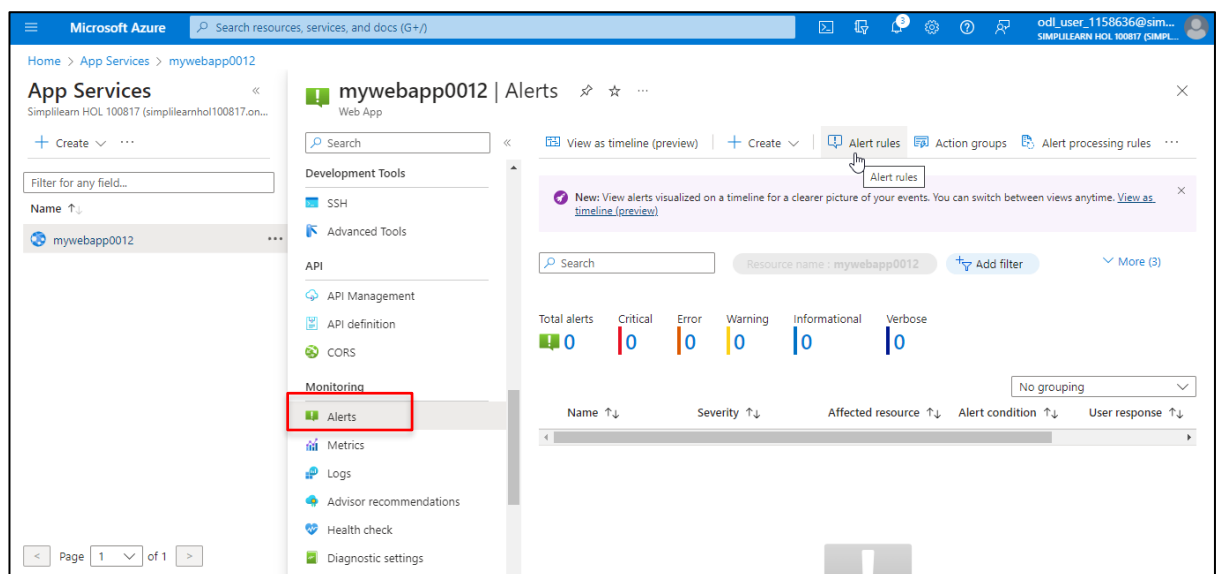
Check every [1 minute](#)

At the bottom, there are two buttons: 'Create' (highlighted with a mouse cursor) and 'Previous'. A tooltip for the 'Create' button shows the text 'Create'.

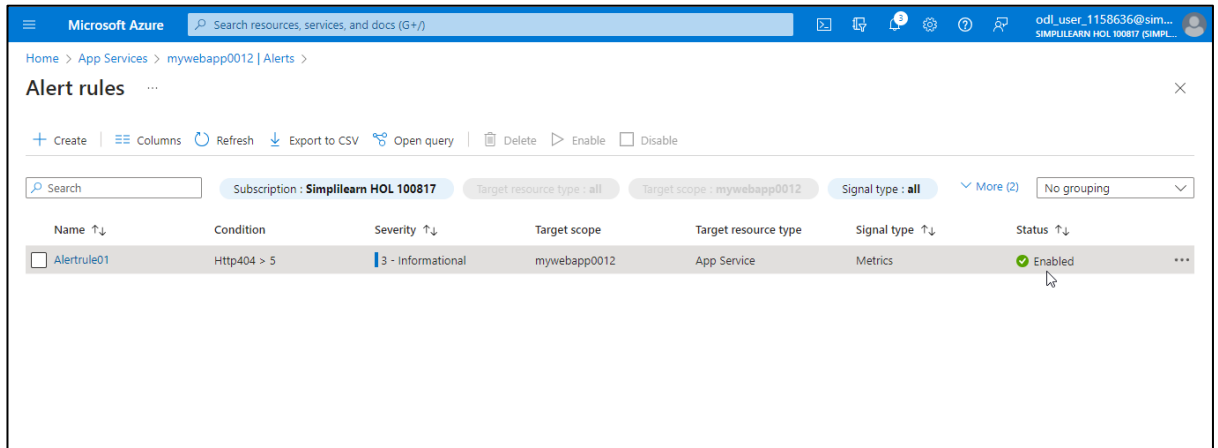
The alert rule is created as shown in the screenshot below:



2.18 On the Alerts page, click on Alert rules



In the **Alert rules** page, you can view the created alert rule **Alertrule01** with the **Status** as **Enabled**:



The screenshot shows the Microsoft Azure portal interface for managing alert rules. The breadcrumb navigation indicates the path: Home > App Services > mywebapp0012 | Alerts >. The page title is "Alert rules". Below the title, there are several action buttons: Create, Columns, Refresh, Export to CSV, Open query, Delete, Enable, and Disable. A search bar is present, and filters for Subscription (Simplilearn HOL 100817), Target resource type (all), Target scope (mywebapp0012), and Signal type (all) are applied. A table lists the alert rules, with one rule named "Alertrule01" shown. The rule's condition is "Http404 > 5", severity is "3 - Informational", target scope is "mywebapp0012", target resource type is "App Service", signal type is "Metrics", and its status is "Enabled".

Name ↑↓	Condition	Severity ↑↓	Target scope	Target resource type	Signal type ↑↓	Status ↑↓
☐ Alertrule01	Http404 > 5	3 - Informational	mywebapp0012	App Service	Metrics	Enabled

By following the above steps, you have successfully created an alert that triggers when the number of HTTP 404 errors for the web app surpasses a threshold of 5, notifying the appropriate stakeholders.